

$^{34}\text{S}(\text{pol d,p})$ 1977Ab07 $J^\pi=0^+$ for ^{34}S g.s.

1977Ab07: A vector-polarized 11.8-MeV deuteron beam was produced from the University of Wisconsin Lamb-shift polarized ion source and tandem accelerator. The target was a 90% enriched ^{34}S , 610- $\mu\text{g}/\text{cm}^2$ thick, evaporated onto a 100- $\mu\text{g}/\text{cm}^2$ gold foil and covered by a 50- $\mu\text{g}/\text{cm}^2$ gold layer. The reaction products were detected using four ΔE -E counter telescopes of freon-cooled surface-barrier and Si(Li) detectors with FWHM $\approx$ 60 keV. Measured angular distributions of cross sections $\sigma(E_p, \theta)$ and angular distributions of analyzing powers $iT_{11}(E_p, \theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from finite-range, non-locality corrected DWUCK-DWBA analysis of $\sigma(E_p, \theta)$ and $iT_{11}(E_p, \theta)$. Also see 1976AbZX.

 ^{35}S Levels

Spectroscopic factor $C^2S = \sigma(\theta)_{\text{exp}} / \sigma(\theta)_{\text{DWBA}} / N$, where $N=1.53$ is a normalization factor (1977Ab07).

E(level) [†]	J^π	L [@]	C^2S [@]	Comments
0	$3/2^+ \ddagger$	2	0.56, 0.48 [‡]	
1572	$1/2^+ \#$	0	0.27, 0.18 [#]	
1992	$7/2^- \#$	3	1.16, 0.73 [#]	
2348	$3/2^- \#$	1	0.56, 0.46 [#]	
2718	$5/2^+ \#$	2	0.03, 0.02 [#]	
2939	$3/2^+, 5/2^+$	2	0.10, 0.06	J^π : 1977Ab07 states that the analyzing power seems to favor $5/2^+$ but they do not think that the evidence is strong enough to support an unambiguous spin-parity assignment. C^2S : extracted from an incoherent superposition of direct and compound cross sections in 1977Ab07.
3421	$5/2^+ \#$	2	0.04, 0.03 [#]	
3563				
3802	$3/2^- \#$	1	0.10, 0.08 [#]	
3818				
4190	$1/2^- \ddagger$	1	0.15, 0.13 [‡]	
4482	$7/2^- \#$	3	0.06, 0.05 [#]	
4575				
4837				
4904	$1/2^- \ddagger$	1	0.48, 0.41 [‡]	
4963	$3/2^- \#$	1	0.19, 0.17 [#]	
5058	$7/2^- \#$	3	0.03, 0.03 [#]	
5126				

[†] As given in 1977Ab07, originally from 1973EnVA rounded to the nearest integer.

[‡] L-1/2 transfer from analyzing power measurements.

[#] L+1/2 transfer from analyzing power measurements.

[@] From DWUCK-DWBA analysis of the measured $\sigma(\theta)$ in 1977Ab07. C^2S calculations use two sets of optical model potentials. Set 1: the deuteron elastic potential and the Becchetti-Greenlees proton parameters; Set 2: Adiabatic model parameters and the Becchetti-Greenlees proton parameters.